National College of Ireland

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Master of Science in Cloud Computing – Full Time – Year 1 - MSCCLOUD1_A

Master of Science in Cloud Computing– Full Time – Year 1 - MSCCLOUD1_B

Semester One Examinations

Wednesday 10th January 2024
2pm – 4pm

Blockchain Concepts and Technologies

Duration of exam: **2 Hours**
Attachments: **None Required**

Answer All Questions

You should answer **only the required number** of questions.
If you answer **more** than the required number of questions you should **cancel** (put a line through) the question(s) you do not want marked.
If you do not cancel extra questions, the examiner will mark the questions **in the order presented** in your answer book until the required number of questions have been marked.

Question 1: (5 marks)

The public Ethereum network uses which Consensus Mechanism (select one response):

1. POW (Proof of Work)
2. POA (Proof of Authority)
3. POS (Proof of Stake)
4. PBFT (Practical Byzantine Fault Tolerance)
5. RAFT (Reliable, Replicated, Redundant, And Fault-Tolerant)

Question 2: (5 marks)

The central novelty of the Bitcoin blockchain was (select the one response that most applies):

1. The concept of linking data block via a hash/pointer to the previous data block (the blockchain)
2. The use of digital signatures to provide cryptographically sound money
3. The use of Proof Of Work to provide decentralized consensus
4. Open sourcing of the underlying code to enable online collaboration

Question 3: (5 marks)

Which of the following are properties of money (select all applicable responses):

1. Medium of Exchange
2. Compliance with Global Financial Regulations
3. Unit of Account
4. Issuance by a Central Bank
5. Store of Value
6. Security of Transfer
7. Resistance to Counterfeiting

Question 4: (5 marks)

A blockchain hard fork is caused by (select one response):

1. A change in the underlying cryptographic protocols
2. A change in governance teams
3. A software change invalidating future blocks generated by previous software version
4. A split in the value of the currency similar to a stock split

Question 5: (5 marks)

What is the purpose of gas in the Ethereum blockchain (select most applicable response):

1. To provide a unit of account for transaction cost
2. To prevent infinite loops in smart contract code
3. As an alternative to transaction fees
4. To prevent chain forks

Question 6: (5 marks)

Transactions on the Bitcoin blockchain are (select one response):

1. Anonymous and untraceable
2. Pseudonymous and traceable
3. Obfuscated by public key cryptography and private by design
4. Available only to registered users authorized by Bitcoin Foundation

Question 7: (5 marks)

Ethereum generates blocks approximately every (select one response):

1. 10 minutes
2. 15 seconds
3. 90 seconds
4. 24 hours

Question 8: (5 Marks)

What is Proof of Stake? Detail your response including references to randomness, its role in decentralized consensus, power consumption and Delegated Proof of Stake.

Question 9 (20 marks)

Define the four fundamental concepts, which you believe made the creation of the bitcoin blockchain possible in 2009, and how these concepts interact and combine to enable blockchain as a disruptive technology. You must also critically appraise why you have selected these as the most fundamental concepts.

Question 10 (10 marks)

Critically explain and detail a transaction lifecycle process, you are free to choose either Bitcoin or Ethereum transaction lifecycle. Within your response ensure to detail each key function and process within the lifecycle, an illustration or diagram can be used to support your response.

Question 11 (5 marks)

Detail why you are expected to wait for a certain number of confirmations after receiving a cryptocurrency transaction?

Question 12 (10 marks)

The European Union in April of 2023 has passed legislation relating to the mining and holding of Crypto-Assets. Provide a brief summary of this legislation.

Question 13 (5 Marks)

Explain the concept of an ICO, within your response highlight a positive and negative impact of such an action.

Question 14 (10 marks)

Briefly Explain the concept of POW Mining in the BTC network.